

Tutorial 2

- 1) Define LTI systems. Explain with examples, how periodic and non-periodic signals are distinguished.
- 2) Define energy and power signal.
- 3) How do you classify the signal?
- 4) Check whether $x(t) = A \cos \omega t$ is a power or energy type signal.
- 5) Determine whether the signal $x(t) = A[u(t+a) - u(t-a)]$ for $a > 0$ is a power or energy signal.
- 6) Find the energy and power of signal $x(t) = e^{-4t} u(t)$ and justify whether it is energy signal or power signal.
- 7) How would you say that a system is BIBO stable? Comment about the stability for the following system: $y(n) = 2x(n+1) + [x(n-1)]^2$
- 8) Comment about the linearity, stability, time-invariance and causality for the following system: $y(t) = t^2 x(t)$
- 9) Check causality and time-invariance of a signal:
$$y(t) = x(t) + u(t-5)$$
- 10) Check whether the given system is linear, causal and time invariant or not.
 - i) $y(t) = \log x(t)$
 - ii) $y(t) = t x(t)$
- 11) Explain why a sinusoidal wave is periodic and deterministic signal.
- 12) Find whether the signal is energy or power signal: $\delta[n]$